Data transformation with dplyr

Day 2 - Introduction to Data Analysis with R

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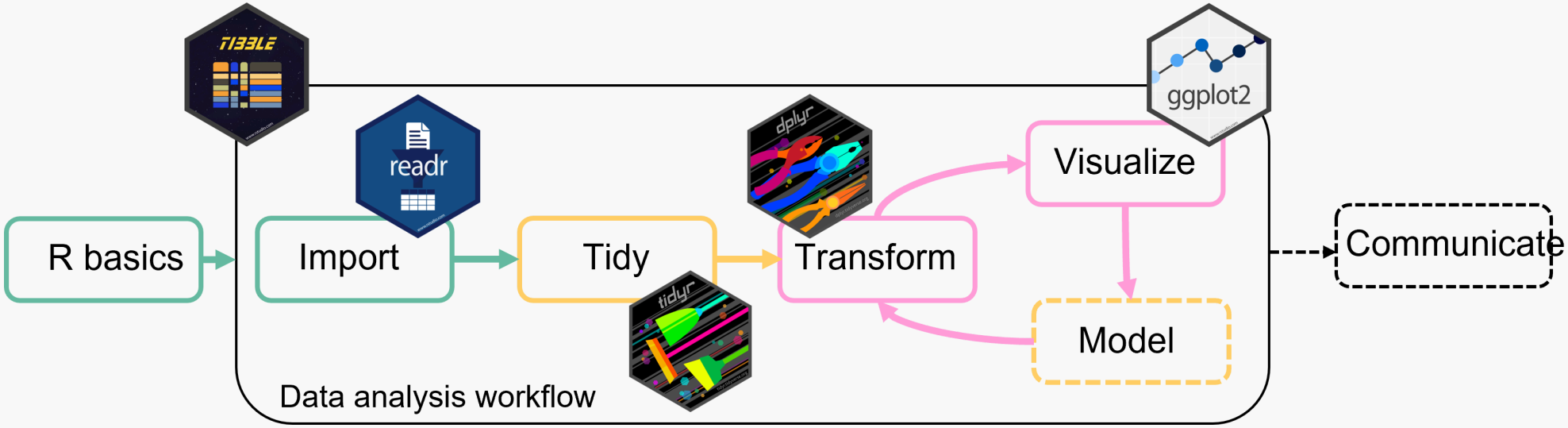
Data transformation

Data transformation is an important step in **understanding** the data and **preparing** it for further analysis.



We can use the tidyverse package `dplyr` for this.

Data transformation

With `dplyr` we can (among other things)

- **Filter** data to analyse only a part of it
- **Create** new variables
- **Summarize** data
- **Combine** multiple tables
- **Rename** variables
- **Reorder** observations or variables

To get started load the package `dplyr`:

```
library(dplyr)
# or
library(tidyverse)
```

Dplyr basic vocabulary

All of the `dplyr` functions work similarly:

- **First argument** is the data (a tibble)
- **Other arguments** specify what to do exactly
- **Return** a tibble

The data

Data set `and_vertebrates` with measurements of a trout and 2 salamander species in different forest sections.

- `year`: observation year
- `section`: CC (clear cut forest) or OG (old growth forest)
- `unittype`: channel classification (C = Cascade, P = Pool, ...)
- `species`: Species measured
- `length_1_mm`: body length [mm]
- `weight_g`: body weight [g]

Coastal giant salamander (terrestrial form) Andrews Forest Program by Lina DiGregorio via CC-BY from https://andrewsforest.oregonstate.edu

References: Kaylor, M.J. and D.R. Warren. (2017) and Gregory, S.V. and I. Arismendi. (2020) as provided in the

The data

Data set `and_vertebrates` with measurements of a trout and 2 salamander species in different forest sections.

```
library(lterdatasampler)
vertebrates <- and_vertebrates |>
  select(year, section, unittype, species, length_1_mm, weight_g) |>
  filter(species != "Cascade torrent salamander")
vertebrates
#> # A tibble: 32,191 × 6
#>     year section unittype species         length_1_mm weight_g
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#>  1  1987 CC      R        Cutthroat trout          58     1.75
#>  2  1987 CC      R        Cutthroat trout          61     1.95
#>  3  1987 CC      R        Cutthroat trout          89     5.6
#>  4  1987 CC      R        Cutthroat trout          58     2.15
#>  5  1987 CC      R        Cutthroat trout          93     6.9
#>  6  1987 CC      R        Cutthroat trout          86     5.9
#>  7  1987 CC      R        Cutthroat trout         107    10.5
#>  8  1987 CC      R        Cutthroat trout         131    20.6
#>  9  1987 CC      R        Cutthroat trout         103     9.55
#> 10  1987 CC      R        Cutthroat trout         117    13
#> # ℹ 32,181 more rows
```

`filter()`

> picks rows based on their value

filter()

Filter only the trout species:

```
filter(vertebrates, species == "Cutthroat trout")
#> # A tibble: 20,433 × 6
#>     year section unittype species         length_1_mm weight_g
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#>  1  1987 CC      R        Cutthroat trout          58     1.75
#>  2  1987 CC      R        Cutthroat trout          61     1.95
#>  3  1987 CC      R        Cutthroat trout          89     5.6
#>  4  1987 CC      R        Cutthroat trout          58     2.15
#>  5  1987 CC      R        Cutthroat trout          93     6.9
#>  6  1987 CC      R        Cutthroat trout          86     5.9
#>  7  1987 CC      R        Cutthroat trout         107    10.5
#>  8  1987 CC      R        Cutthroat trout         131    20.6
#>  9  1987 CC      R        Cutthroat trout         103     9.55
#> 10  1987 CC      R        Cutthroat trout         117    13
#> # ℹ 20,423 more rows
```

`filter()` goes through each row of the data and return only those rows where the value for `species` is `"Cutthroat trout"`

filter()

You can also combine filters using logical operators (&, |, !):

```
filter(vertebrates, species == "Cutthroat trout" & year == 1987)
#> # A tibble: 603 × 6
#>     year section unittype species         length_1_mm weight_g
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#>  1  1987 CC      R        Cutthroat trout          58     1.75
#>  2  1987 CC      R        Cutthroat trout          61     1.95
#>  3  1987 CC      R        Cutthroat trout          89     5.6
#>  4  1987 CC      R        Cutthroat trout          58     2.15
#>  5  1987 CC      R        Cutthroat trout          93     6.9
#>  6  1987 CC      R        Cutthroat trout          86     5.9
#>  7  1987 CC      R        Cutthroat trout         107    10.5
#>  8  1987 CC      R        Cutthroat trout         131    20.6
#>  9  1987 CC      R        Cutthroat trout         103     9.55
#> 10  1987 CC      R        Cutthroat trout         117    13
#> # ℹ 593 more rows
```

filter() + %in%

Use the %in% operator to filter rows based on multiple values, e.g. unittypes

```
unittype_select <- c("R", "C", "S")
filter(vertebrates, unittype %in% unittype_select)
#> # A tibble: 19,619 × 6
#>     year section unittype species         length_1_mm weight_g
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#>  1  1987 CC      R        Cutthroat trout          58     1.75
#>  2  1987 CC      R        Cutthroat trout          61     1.95
#>  3  1987 CC      R        Cutthroat trout          89     5.6
#>  4  1987 CC      R        Cutthroat trout          58     2.15
#>  5  1987 CC      R        Cutthroat trout          93     6.9
#>  6  1987 CC      R        Cutthroat trout          86     5.9
#>  7  1987 CC      R        Cutthroat trout         107    10.5
#>  8  1987 CC      R        Cutthroat trout         131    20.6
#>  9  1987 CC      R        Cutthroat trout         103     9.55
#> 10  1987 CC      R        Cutthroat trout         117    13
#> # ℹ 19,609 more rows
```

filter() + is.na()

Filter only rows that don't have a value for the weight

```
filter(vertebrates, is.na(weight_g))
#> # A tibble: 13,259 × 6
#>     year section unittype species         length_1_mm weight_g
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#>  1  1993 CC      P        Cutthroat trout          93       NA
#>  2  1993 CC      P        Cutthroat trout         175       NA
#>  3  1993 CC      P        Cutthroat trout         104       NA
#>  4  1993 CC      P        Cutthroat trout          98       NA
#>  5  1993 CC      P        Cutthroat trout          97       NA
#>  6  1993 CC      P        Cutthroat trout         123       NA
#>  7  1993 CC      P        Cutthroat trout         149       NA
#>  8  1993 CC      P        Cutthroat trout         100       NA
#>  9  1993 CC      P        Cutthroat trout         118       NA
#> 10  1993 CC      P        Cutthroat trout         163       NA
#> # ℹ 13,249 more rows
```

Or the opposite: filter only the rows that have a value for the weight

```
filter(vertebrates, !is.na(weight_g))
```

filter() + between()

Combine different filters:

Filter rows where the value for `year` is between 2000 and 2005

```
filter(vertebrates, between(year, 2000, 2005))
#> # A tibble: 6,662 × 6
#>     year section unittype species         length_1_mm weight_g
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#>  1  2000 CC      C        Cutthroat trout          84       NA
#>  2  2000 CC      C        Cutthroat trout         132       NA
#>  3  2000 CC      C        Cutthroat trout         105       NA
#>  4  2000 CC      C        Cutthroat trout          41       NA
#>  5  2000 CC      C        Cutthroat trout          42       NA
#>  6  2000 CC      C        Cutthroat trout          42       NA
#>  7  2000 CC      C        Cutthroat trout          41       NA
#>  8  2000 CC      C        Cutthroat trout          51       NA
#>  9  2000 CC      C        Cutthroat trout          45       NA
#> 10  2000 CC      C        Cutthroat trout          44       NA
#> # ℹ 6,652 more rows
```

Or you could also do it like this:

```
filter(vertebrates, year >= 2000 & year <= 2005)
```

Useful `filter()` helpers

These functions and operators help you filter your observations:

- relational operators `<`, `>`, `==`, …
- logical operators `&`, `|`, `!`
- `%in%` to filter multiple values
- `is.na()` to filter missing values
- `between()` to filter values that are between an upper and lower boundary
- `near()` to compare floating points (use instead of `==` for doubles)

`select()`

> picks columns based on their names

select()

Select the columns species, length_1_mm, and year

```
select(vertebrates, species, length_1_mm, year)
#> # A tibble: 32,191 × 3
#>    species         length_1_mm  year
#>    <chr>                 <dbl> <dbl>
#>  1 Cutthroat trout          58  1987
#>  2 Cutthroat trout          61  1987
#>  3 Cutthroat trout          89  1987
#>  4 Cutthroat trout          58  1987
#>  5 Cutthroat trout          93  1987
#>  6 Cutthroat trout          86  1987
#>  7 Cutthroat trout         107  1987
#>  8 Cutthroat trout         131  1987
#>  9 Cutthroat trout         103  1987
#> 10 Cutthroat trout         117  1987
#> # ℹ 32,181 more rows
```

Remove variables using -

```
select(vertebrates, -species, -length_1_mm, -year)
```

select() + starts_with()

Select all columns that start with "s"

```
select(vertebrates, starts_with("s"))
```

```
#> # A tibble: 32,191 × 2
#>   section species
#>   <chr>   <chr>
#> 1 CC      Cutthroat trout
#> 2 CC      Cutthroat trout
#> 3 CC      Cutthroat trout
#> # ℹ 32,188 more rows
```

You can use the same structure for ends_with() and contains().

```
# this does not make sense for the example data
# but combinations like this are helpful for research data
select(vertebrates, starts_with("_location1"))

select(vertebrates, contains("_id_"))
```

Useful `select()` helpers

- `starts_with()` and `ends_with()`: variable names that start/end with a specific string
- `contains()`: variable names that contain a specific string
- `matches()`: variable names that match a regular expression
- `any_of()` and `all_of()`: variables that are contained in a character vector

`mutate()`

> Adds new columns to your data

mutate()

New columns can be added based on values from other columns

```
mutate(vertebrates, weight_kg = weight_g / 1000)
```

```
#> # A tibble: 32,191 × 7
#>    year section unittype species         length_1_mm weight_g weight_kg
#>   <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>     <dbl>
#> 1  1987 CC      R        Cutthroat trout          58     1.75   0.00175
#> 2  1987 CC      R        Cutthroat trout          61     1.95   0.00195
#> 3  1987 CC      R        Cutthroat trout          89     5.6    0.0056
#> # ℹ 32,188 more rows
```

Add multiple new columns at once:

```
mutate(vertebrates, weight_kg = weight_g / 1000, length_m = length_1_mm / 1000)
```

mutate() + case_when()

Use `case_when` to add column values conditional on other columns.

`case_when()` can combine many cases into one.

```
mutate(
  vertebrates,
  type = case_when(
    species == "Cutthroat trout" ~ "Fish", # case 1
    species == "Coastal giant salamander" ~ "Amphibian", # case 2
    .default = NA # all other
  )
)
#> # A tibble: 32,191 × 7
#>     year section unittype species         length_1_mm weight_g type
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl> <chr>
#>  1  1987 CC      R        Cutthroat trout          58     1.75 Fish
#>  2  1987 CC      R        Cutthroat trout          61     1.95 Fish
#>  3  1987 CC      R        Cutthroat trout          89     5.6  Fish
#>  4  1987 CC      R        Cutthroat trout          58     2.15 Fish
#>  5  1987 CC      R        Cutthroat trout          93     6.9  Fish
#>  6  1987 CC      R        Cutthroat trout          86     5.9  Fish
#>  7  1987 CC      R        Cutthroat trout         107    10.5  Fish
#>  8  1987 CC      R        Cutthroat trout         131    20.6  Fish
#>  9  1987 CC      R        Cutthroat trout         103     9.55 Fish
#> 10  1987 CC      R        Cutthroat trout         117    13    Fish
#> # ℹ 32,181 more rows
```

`summarize()`

> summarizes data

summarize()

summarize will collapse the data to a single row

```
summarize(
  vertebrates,
  mean_length = mean(length_1_mm, na.rm = TRUE),
  mean_weight = mean(weight_g, na.rm = TRUE)
)
#> # A tibble: 1 × 2
#>   mean_length mean_weight
#>         <dbl>       <dbl>
#> 1        73.8        8.91
```

`summarize()` by group

`summarize` is much more useful in combination with the grouping argument `.by`

- **summary** will be calculated **separately for each group**

```
# summarize the grouped data
summarize(
  vertebrates,
  mean_length = mean(length_1_mm, na.rm = TRUE),
  mean_weight = mean(weight_g, na.rm = TRUE),
  .by = species
)
#> # A tibble: 2 × 3
#>   species                  mean_length mean_weight
#>   <chr>                          <dbl>       <dbl>
#> 1 Cutthroat trout                 83.5        8.84
#> 2 Coastal giant salamander        57.0        9.03
```

- Combine variables if you want to summarize by more than one group (e.g. `.by = c(species, unittype)`)

count()

Counts observations by group

```
# count rows grouped by year
count(vertebrates, year)
#> # A tibble: 33 × 2
#>     year     n
#>    <dbl> <int>
#>  1  1987   603
#>  2  1988   302
#>  3  1989   308
#>  4  1990   513
#>  5  1991   626
#>  6  1992   616
#>  7  1993   870
#>  8  1994   948
#>  9  1995   583
#> 10  1996   928
#> # ℹ 23 more rows
```

The pipe |>

> Combine multiple data operations into one command

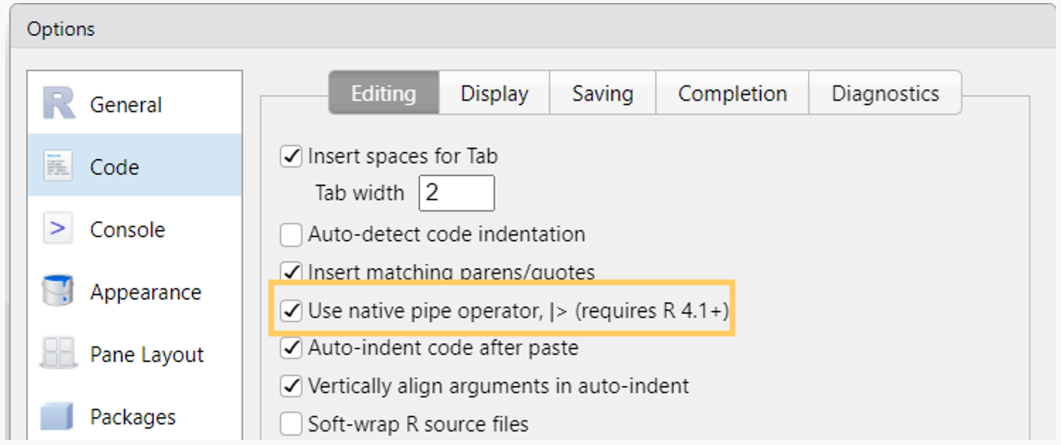
The pipe |>

Data transformation often requires **multiple operations** in sequence.

The pipe operator |> helps to keep these operations clear and readable.

- You may also see `%>%` from the `magrittr` package

Turn on the native R pipe |> in **Tools -> Global Options -> Code**



The pipe |>

Let's look at an example without pipe:

```
# 1: filter rows that have don't have NA in the unittype column
vertebrates_new <- filter(vertebrates, !is.na(unittype))

# 2: summarize mean values by year
vertebrates_new <- count(vertebrates_new, year, species, section)
```

How could we make this more efficient?

Use one **nested function** without intermediate results:

```
vertebrates_new <- count(
  filter(vertebrates, !is.na(unittype)),
  year,
  species,
  section
)
```

But this gets complicated and error prone very quickly

The pipe |>

The pipe operator makes it very easy to combine multiple operations:

```
vertebrates_new <- vertebrates |>
  filter(!is.na(unittype)) |>
  count(year, species, section)

vertebrates_new
```

You can read from top to bottom and interpret the |> as an "and then do".

The pipe |>

But what is happening?

The pipe is "pushing" the result of one line into the first argument of the function from the next line.

```
vertebrates |>
  count(year)

# instead of
count(vertebrates, year)
```

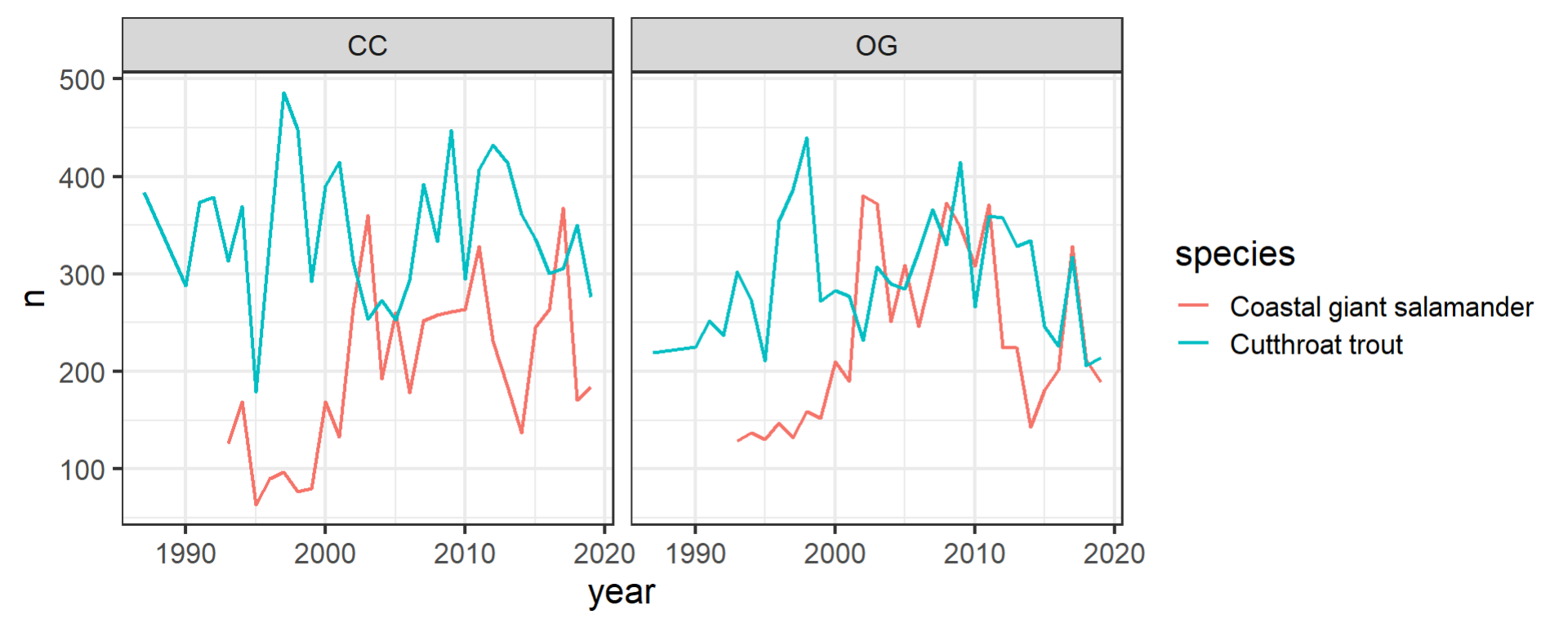
Piping works perfectly with the `tidyverse` functions because they are designed to return a tibble **and** take a tibble as first argument.

> Tip
>
> Use the keyboard shortcut `Ctrl/Cmd + Shift + M` to insert `|>`

The pipe |>

Piping also works well together with `ggplot`

```
vertebrates |>
  filter(!is.na(unittype)) |>
  count(year, species, section) |>
  ggplot(aes(x = year, y = n, color = species)) +
  geom_line() +
  facet_wrap(~section)
```



Combining mulitiple tables

Combine two tibbles by row `bind_rows`

Situation: Two (or more) `tibbles` with the same variables (column names)

```
tbl_a <- vertebrates[1:2, ] # first two rows
tbl_b <- vertebrates[2:nrow(vertebrates), ] # the rest
```

```
tbl_a
```

```
#> # A tibble: 2 × 6
#>    year section unittype species         length_1_mm weight_g
#>   <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#> 1  1987 CC      R        Cutthroat trout          58     1.75
#> 2  1987 CC      R        Cutthroat trout          61     1.95
```

```
tbl_b
```

```
#> # A tibble: 32,190 × 6
#>    year section unittype species         length_1_mm weight_g
#>   <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#> 1  1987 CC      R        Cutthroat trout          61     1.95
#> 2  1987 CC      R        Cutthroat trout          89     5.6
#> # ℹ 32,188 more rows
```

Combine two tibbles by row `bind_rows`

Bind the rows together with `bind_rows()`:

```
bind_rows(tbl_a, tbl_b)
```

```
#> # A tibble: 32,192 × 6
#>    year section unittype species         length_1_mm weight_g
#>   <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#> 1  1987 CC      R        Cutthroat trout          58     1.75
#> 2  1987 CC      R        Cutthroat trout          61     1.95
#> # ℹ 32,190 more rows
```

You can also add an ID-column to indicate which line belonged to which table:

```
bind_rows(a = tbl_a, b = tbl_b, .id = "id")
```

```
#> # A tibble: 32,192 × 7
#>   id     year section unittype species         length_1_mm weight_g
#>   <chr> <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#> 1 a      1987 CC      R        Cutthroat trout          58     1.75
#> 2 a      1987 CC      R        Cutthroat trout          61     1.95
#> 3 b      1987 CC      R        Cutthroat trout          61     1.95
#> # ℹ 32,189 more rows
```

You can use `bind_rows()` to bind as many tables as you want:

```
bind_rows(a = tbl_a, b = tbl_b, c = tbl_c, ..., .id = "id")
```

Join tibbles with `left_join()`

Situation: Two tables that share some but not all columns.

```
vertebrates
```

```
#> # A tibble: 32,191 × 6
#>    year section unittype species         length_1_mm weight_g
#>   <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl>
#> 1  1987 CC      R        Cutthroat trout          58     1.75
#> 2  1987 CC      R        Cutthroat trout          61     1.95
#> # ℹ 32,189 more rows
```

```
# table with more information on the species
species
#> # A tibble: 2 × 2
#>   species                 type
#>   <chr>                   <chr>
#> 1 Cutthroat trout         Fish
#> 2 Coastal giant salamander Amphibian
```

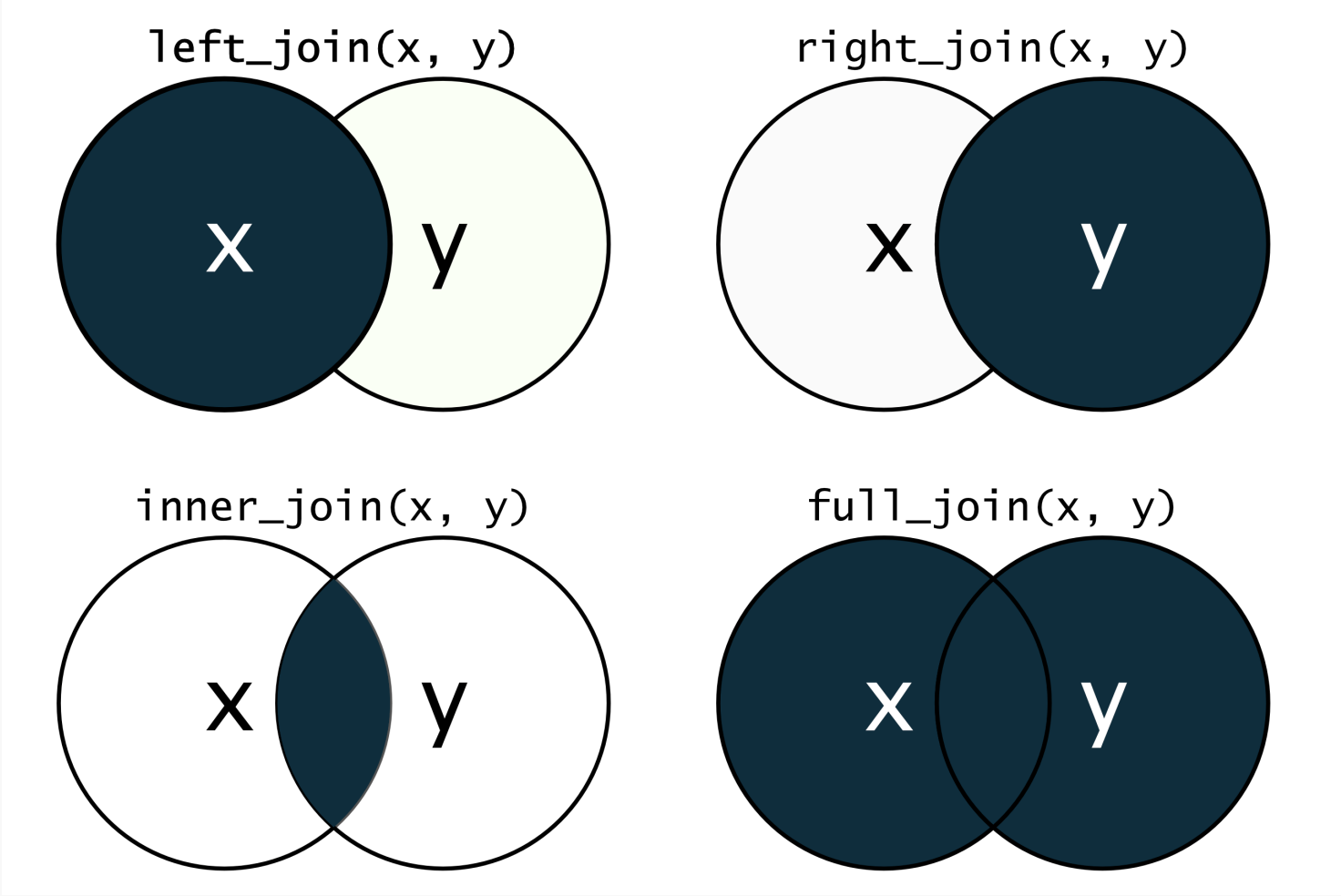
Join tibbles with `left_join()`

Join the two tables by the common column `species`

```
left_join(vertebrates, species, by = "species")
#> # A tibble: 32,191 × 7
#>     year section unittype species         length_1_mm weight_g type
#>    <dbl> <chr>   <chr>    <chr>                 <dbl>    <dbl> <chr>
#>  1  1987 CC      R        Cutthroat trout          58     1.75 Fish
#>  2  1987 CC      R        Cutthroat trout          61     1.95 Fish
#>  3  1987 CC      R        Cutthroat trout          89     5.6  Fish
#>  4  1987 CC      R        Cutthroat trout          58     2.15 Fish
#>  5  1987 CC      R        Cutthroat trout          93     6.9  Fish
#>  6  1987 CC      R        Cutthroat trout          86     5.9  Fish
#>  7  1987 CC      R        Cutthroat trout         107    10.5  Fish
#>  8  1987 CC      R        Cutthroat trout         131    20.6  Fish
#>  9  1987 CC      R        Cutthroat trout         103     9.55 Fish
#> 10  1987 CC      R        Cutthroat trout         117    13    Fish
#> # ℹ 32,181 more rows
```

`left_join()` means that the resulting tibble will contain all rows of `vertebrates`, but not necessarily all rows of `species` (in this case it does though).

Different `*_join()` functions



Summary

> Data transformation with dplyr

Summary I

All `dplyr` functions take a tibble as first argument and return a tibble.

`filter()`

- **pick rows** with helpers
  - relational and logical operators
  - `%in%`
  - `is.na()`
  - `between()`
  - `near()`

Summary II

All `dplyr` functions take a tibble as first argument and return a tibble.

`select()`

- pick columns with helpers
  - `starts_with()`, `ends_with()`
  - `contains()`
  - `matches()`
  - `any_of()`, `all_of()`

Summary III

`arrange()`

- **change order** of rows (adscending)
  - or descending with `desc()`

`mutate()`

- **add columns** but keep all columns
  - `case_when()` for conditional values

Summary IV

`summarize()`

- **collapse rows** into one row by some summary
  - use `.by` argument to summarize by group

`count`

- **count rows** based on a group

Summary V

`bind_rows()`

- **combine rows** of multiple tibbles into one
  - the tibbles need to have the same columns
  - add an id column with the argument `.id = "id"`
  - function `bind_cols()` works similarly just for columns

`left_join()`

- **combine tables** based on common columns

Now you

Task (30 min)

Transform the penguin data set

Find the task description here